Agricultural Soils Management

Land Sciences

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Agricultural Soils Management (A08623)

Short Title:	Agricultural Soils Management		
Faculty	Science and Computing		
Department	Land Sciences		
Cluster	Agriculture		
Author	VNYHAN		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module is designed to equip students with a introductory understanding of the physical, chemical and biological properties of soil. The student will also gain an appreciation of what governs soil quality, and an insight into soil degradation processes, including how these can be prevented and corrected. Students will learn about the macronutrients and micronutrients involved in plant nutrition, including the preparation of a farm nutrient management plan. The student will practice soil sampling and conduct a range of both lab and field-based physical, chemical and biological practical analysis of soils, culminating in the identification of a given soil profile in situ.

Programmes

`AGRI-0102`	BSc (Hons) in Agricultural Science (`WD_SAGRI_B`)	2 / 3 / M
`AGRI-0102`	BSc in Agriculture (`WD_SAGUL_D`)	2 / 3 / M

Indicative Content

- Soil formation and composition
- Soil physical properties: texture and structure
- Soil chemical properties: pH, CEC and plant nutrition
- Soil biological properties: organic matter and ecology
- Soil compaction, erosion and drainage
- Soil classification
- Soil sampling and analysis
- Nutrient management planning

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain the processes involved in soil formation.
2. Describe soil composition and soil classification, correlate to soil functionality.
3. Discuss the soil properties of texture and structure, and relate to soil degradation processes and soil management.
4. Describe the function of key plant nutrients, their movement and availability in soils, and relate to management of organic and inorganic fertilisers.
5. Classify and describe soil organic matter and soil biological organisms, and relate to soil quality and management.
6. Complete a basic soil analysis to identify a given soil profile and defend findings.
7. Complete a whole farm nutrient management plan.
8. Use a range of laboratory techniques to characterise soils.

Learning and Teaching Methods

- Lectures.
- Laboratory practicals.
- Site visits.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5
Continuous Assessment	50%	
In-Class Assessment	20%	7
Practical	30%	6,8

Assessment Criteria

<40%: Inadequate understanding of the fundamentals of the learning outcomes and insufficient competence in soil management.

40%-49%: Will be able to show a basic knowledge of the key learning outcomes including soil physical, chemical and biological properties as well as be able to perform basic soil analysis.

50%-59%: As well as the above, the student will be able to demonstrate an understanding of some of the complexities of soil management.

60%-69%: In addition to the above, the student will demonstrate a more detailed knowledge of soil management and a high level of competence in the area.

70%-100%: All of the above to an excellent level. The above advanced knowledge of the topic will be supported by superior skills.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- "Soil and soil fertility." http://www.teagasc.ie/soil/
- Brady, N.C. and R.R. Weil. _The Nature and Properties of Soils 14 (ed)_. Upper Saddle River, NJ: Prentice-Hall, 2008.
- Environment, Agency. _Think soils assessment to avoid erosion and runoff_. Bristol: Environment Agency, 2008.
- Moore, M (editor). _Teagasc manual on drainage and soil management_. Oakpark: Teagasc, 2013.

Supplementary Material

- Breen, J. and G. Mullen. _Agicultural Science_. Dublin: Folens, 1992.
- Foth, H.D. _Fundamentals of So__il Science_. New York: John Wiley & Sons, 1990.

- Jeffery, S., C. Gardi, A. Jones, L. Montanarella, L. Marmo and Miko, L., Ritz, K., Peres, G., Rombke, J. and van der Putten, H, (eds). _European Atlas of soil biodiversity_. Luxembourg: Publications office of the EU, 2010.
- Westerman,, R.L. (editor). _Soil T__esting and Plant Analysis_. Madison, Wisconsin: Soil Science Society of America, 1996.

Requested Resources

- Room Type: Computer Lab
- Room Type: Biology Lab